

Question 1

2 marks 101

Pierre pushes his car into a garage. The radius of a tire on his car is 22 cm. Determine the distance travelled by his car if the tire rotated a total of 1000° .

Question 2

2 marks 102

Solve, algebraically.

$$7^{\frac{x}{2}} = 85$$

Question 3

4 marks 103

Solve, algebraically, over the interval $[0, 2\pi)$.

$$\sin x(\sec x + 3) = 0$$

Question 4

3 marks 104

Brahim invests \$2500 at an annual interest rate of 6.75% compounded monthly. Determine, algebraically, how many years it will take for his investment to reach an amount of \$10 500.

Use the formula:

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

where A = the amount of the investment after t years

P = the principal of the investment

r = the annual interest rate (as a decimal)

n = the number of compounding periods per year

t = the length of the investment in years

Question 5



2 marks

105

There are 13 adults and 18 children who can be selected to go on a trip. Determine the number of ways 4 adults and 7 children can be selected if Sandra, one of the adults, must be selected.

In the binomial expansion of $\left(\frac{2}{x^2} - x^3\right)^9$, determine and simplify the 6th term.

Note: A calculator is not required for the remaining test questions.

Question 7

1 mark 107

Given that $f(x) = \{(-1, 0), (0, 2), (1, -3), (2, 4)\}$, evaluate $f(f(0))$.

Question 8

1 mark 108

The point $\left(-\frac{5}{6}, b\right)$ is on the unit circle and is in quadrant III.

Determine the exact value of b .

Question 9

1 mark 109

Given the following row of Pascal's Triangle, determine the values of the next row.

1 6 15 20 15 6 1

Question 10

3 marks

110

The following transformations are applied to $f(x)$, resulting in a new function, $g(x)$.

- reflection over the x -axis
- vertical stretch by a factor of 3
- horizontal stretch by a factor of 4

State the equation of $g(x)$ in terms of $f(x)$.

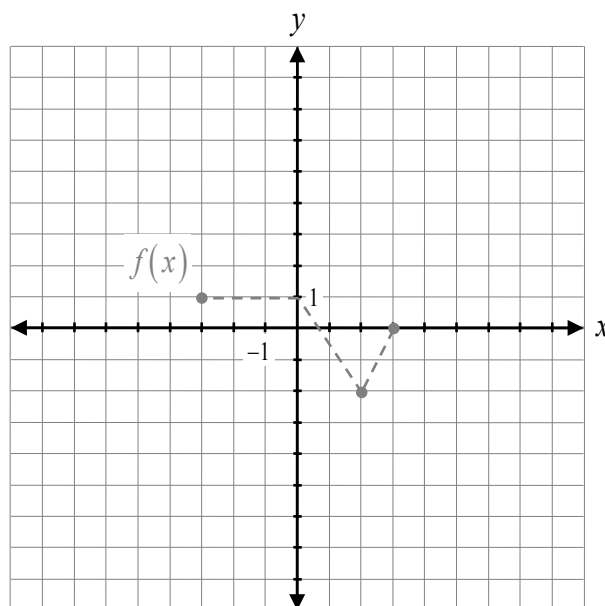
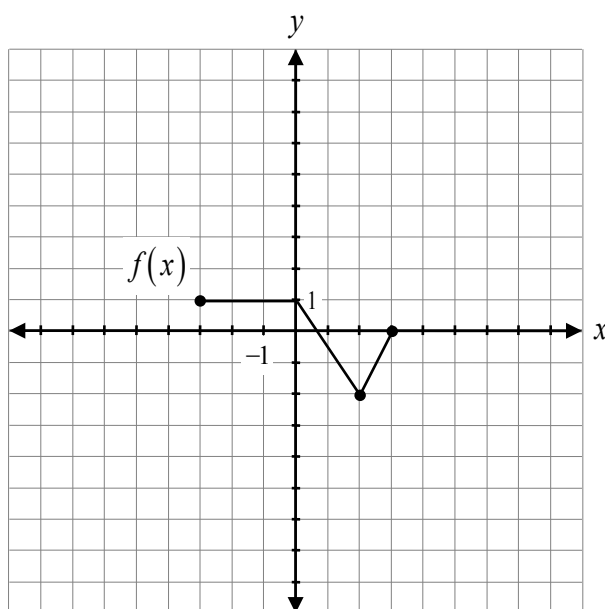
$g(x) =$ _____

Question 11

3 marks

111

Given the graph of $f(x)$, sketch the graph of $y + 1 = 2f(x - 3)$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$

Question 12

1 mark 112

State the equation of the horizontal asymptote of $f(x) = \frac{2x^2 - 3x + 5}{4x^2 + 2x - 7}$.

Question 13

2 marks

113

Given that $(x + 4)$ is one of the factors of $p(x) = x^3 + 6x^2 - 32$, express $p(x)$ in completely factored form.

$p(x) =$ _____

Question 14

3 marks

114

Prove the identity for all permissible values of θ .

$$\frac{2\cos^2\theta}{1-\cot\theta} = \frac{\sin 2\theta}{\tan\theta - 1}$$

Left-Hand Side	Right-Hand Side

Question 15

1 mark 115

Restaurant A has 5 types of hamburgers, 2 types of french fries, and 10 types of drinks.

Restaurant B has 4 types of hamburgers, 5 types of french fries, and 6 types of drinks.

If a meal is made up of a hamburger, french fries, and a drink, justify which restaurant offers a greater variety of meals.

Question 16

2 marks

116

Express $\log_7(2x - 5) + 2\log_7 3$ as a single logarithm.

Question 17

1 mark 117

Explain why $11!$ is not the total number of 11-letter arrangements that can be made from the word CELEBRATION.

No marks will be awarded for work done on this page.

